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SYSTEMATIC REVIEW

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Return-to-sport rate and time in elite athletes after ankle syndesmosis injuries: a systematic review and meta-analysis

Ran Li[1], Changlu Sun[2], Pengkun Qi[3*] and Zhonghao Xu[2*]

Abstract

Background Ankle syndesmosis injuries are a significant concern in elite athletes, often resulting in prolonged recovery and uncertainty regarding optimal management. Although most athletes eventually return to sport (RTS), reported RTS rates and timelines vary widely due to differences in treatment strategies. Given this heterogeneity and lack of consensus, this study systematically reviews RTS rates and time to RTS in elite athletes following syndesmotic ankle injuries.

Methods A systematic search of five databases was performed through September 2025 to identify studies on RTS outcomes in elite athletes with ankle syndesmosis injuries. Pooled RTS proportions were calculated using a random-effects model with logit transformation, and time outcomes were synthesized using random-effects models or descriptive methods. Subgroup, sensitivity, and meta-regression analyses were conducted to assess heterogeneity. Study quality was evaluated using the Newcastle–Ottawa Scale, and publication bias was assessed with funnel plots and Egger's test.

Results Fourteen studies comprising 901 elite athletes were included. The pooled RTS rate following ankle syndesmosis injuries was 96% (95% CI 93–98%), with low-to-moderate heterogeneity ($I^2 = 27\%$). Subgroup analysis showed comparable RTS rates for suture-button fixation (98%) and nonoperative management (98%). Sensitivity analyses confirmed the robustness of the findings. Publication bias was suggested by funnel plot asymmetry and Egger's test ($p = 0.0002$). The average time to RTS across studies was approximately 58 days.

Conclusion Elite athletes sustaining ankle syndesmosis injuries demonstrate a high likelihood of returning to sport, with a pooled RTS rate of 96%. Comparable outcomes were observed following both suture-button fixation and nonoperative management, with most athletes resuming play within 2 months. Most athletes can anticipate favorable outcomes regardless of treatment strategy, though management should still be individualized based on injury severity and sport-specific demands.

Keywords Ankle syndesmosis injury, High ankle sprain, Elite athletes, Return to sport, Rehabilitation

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Background

Ankle syndesmosis injuries, commonly referred to as high ankle sprains, represent a clinically significant problem in elite athletes. Although less frequent than lateral ligament sprains, they account for a considerable proportion of ankle injuries in high-level sport [1]. Epidemiological studies estimate that syndesmosis injuries comprise 10–18% of all ankle sprains in the general population, with an even greater prevalence among professional athletes [2]. These injuries typically occur in contact sports such as rugby and soccer, most often through forced external rotation or dorsiflexion mechanisms, and are associated with greater morbidity and prolonged absence from play compared with lateral ankle sprains [3]. Recovery is frequently extended, with athletes often requiring a minimum of six to eight weeks before resuming activity [4]. Beyond the protracted recovery, syndesmotic injuries carry the risk of persistent disability, uncertainty regarding treatment, and the potential to affect career longevity [5], with inadequately recognized or untreated instability placing patients at increased risk of chronic pain and post-traumatic ankle osteoarthritis [6].

Management strategies for syndesmosis injuries vary depending on severity [3]. Stable or low-grade injuries are typically managed conservatively with immobilization, bracing, and structured rehabilitation, whereas unstable or high-grade injuries, including those with diastasis or complete ligament disruption, generally require surgical fixation [7]. Obvious diastasis typically needs to be reduced and fixed operatively, whereas the optimal management of less severe injuries remains more controversial; nonoperative treatment may be effective but often entails a prolonged period of rehabilitation, and in professional athletes, more aggressive surgical treatment is frequently warranted to facilitate a reliable return to play [8]. Rigid screw fixation remains the traditional standard, though dynamic suture-button devices have gained popularity for their potential to preserve physiologic motion and facilitate earlier return to competition [9]. This trend is supported by a recent meta-analysis, which found that suture-button fixation achieved more favorable outcomes than syndesmotic screw fixation in ankle syndesmosis injuries [10]. Regardless of treatment pathway, rehabilitation protocols focus on restoring joint stability, neuromuscular control, and sport-specific function to enable safe return to sport.

Although numerous investigations have reported RTS rates and timelines following syndesmotic injuries, findings remain inconsistent. Previous systematic reviews suggest that between 94 and 99% of elite athletes ultimately return to play [11–13]. However, the time required to resume full participation varies widely. Nonoperative cases often return within 6–8 weeks, while surgically treated athletes may require 10–12 weeks or longer [12, 14]. More recent studies in professional sport further highlight variability, with return times ranging from 6–7 weeks to as long as three to four months, depending on injury severity, concomitant pathology, and sport-specific demands [15]. Such heterogeneity likely reflects differences in injury grading, associated lesions such as fractures or cartilage damage, and variation in surgical or rehabilitation protocols [14].

Given the wide variability in RTS outcomes and the absence of consensus on optimal management, a systematic synthesis of the evidence is needed. The present systematic review and meta-analysis aims to provide an updated and comprehensive summary of RTS rates and timelines in elite athletes following ankle syndesmosis injuries, thereby offering clearer guidance for clinicians, athletes, and decision-makers in the sporting context.

Methods

This review was designed and reported in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [16]. The study protocol was prospectively submitted to the PROSPERO international prospective register of systematic reviews, where it was accepted on September 16, 2025 (registration ID: CRD420251149156).

Search strategy and selection

We systematically searched PubMed, Embase, Web of Science, and the Cochrane Library from their inception to September 2025. The search strategy combined terms related to athletes (e.g., athlete, elite athlete, professional athlete), ankle syndesmosis injuries (e.g., syndesmosis injury, high ankle sprain, ankle syndesmosis), and outcomes of interest (e.g., return to sport, return to play, time to return, recovery time). Boolean operators (AND and OR) were used to combine these terms, and appropriate subject headings (e.g., MeSH in PubMed) were applied when available; full search strategies are provided in Supplementary File 1.

To ensure that potentially relevant records were not missed, the reference lists of previous systematic reviews and all included studies were hand-searched, and forward citation tracking was conducted through Web of Science. No restrictions were imposed on language, year of publication, or geographical region, and both published and in-press articles were considered eligible. Duplicate records were identified and removed using EndNote software, after which titles and abstracts were screened for relevance. Full-text screening of potentially eligible articles was subsequently performed against the prespecified eligibility criteria. At both stages, two reviewers (R.L. and C.S.) independently assessed each record, with disagreements resolved through consensus or consultation with a third reviewer (Z.X.).

Eligibility criteria

We included studies that investigated elite or professional athletes who sustained ankle syndesmosis injuries, confirmed either clinically or through imaging or surgical findings, and managed through surgical or non-surgical approaches. Eligible studies were required to report at least one outcome related to sport resumption, including return-to-sport rates, the time needed to return to competition, or recovery duration expressed in days, weeks, or months. Both prospective and retrospective cohort studies, as well as randomized controlled trials and case series, were considered for inclusion. No restrictions were imposed on publication language, year, or geographic region, ensuring a comprehensive synthesis of available evidence.

We excluded studies that consisted solely of narrative or expert reviews, opinion pieces, editorials, or conference abstracts that did not provide extractable data. Likewise, studies focusing on recreational or non-elite athletes, or those combining multiple lower limb injuries without presenting separate data for syndesmosis injuries, were not eligible. To maximize completeness, the reference lists of all relevant systematic reviews and included articles were also screened to identify additional eligible records.

Data from each included study were extracted independently by R.L. and C.S. using a standardized form, and all discrepancies were resolved through consultation with the corresponding author (Z.X. and P.Q.).

Data extraction

Data from each included study were extracted independently by R.L. and C. S. using a standardized form, and all discrepancies were resolved through consultation with the corresponding authors (Z.X. and P.Q.). The following information was recorded: author, year of publication, country, study design, and sample size; participant characteristics such as sport discipline, competition level, mean age, and sex distribution; and injury- and treatment-related details, including diagnostic methods, type of syndesmosis injury, and management strategy (surgical or conservative).

For the primary analysis, data on return-to-sport rates and the time to sport resumption were extracted, along with the specific definitions of RTS used in each study and the duration of follow-up. When outcomes were reported as proportions, the number of athletes who successfully returned to sport and the total number of participants were extracted, and corresponding rates were calculated. For time-related outcomes, we collected the mean and standard deviation (SD) of time to RTS whenever available. If only median and range or interquartile range were reported, these were converted to approximate means and SDs using established statistical methods. In cases where relevant outcomes were presented exclusively in graphical form, data were estimated using digital extraction software. When essential information was missing or unclear, study authors were contacted to obtain additional details.

Publication bias and sensitivity analysis

To evaluate the risk of publication bias, funnel plots were visually inspected when the number of eligible studies for an outcome was sufficient, and asymmetry was further examined using Egger's regression test [17]. Sensitivity analyses were performed to determine the stability of the pooled results. This involved repeating the meta-analysis after omitting one study at a time and comparing the consistency of the effect estimates [18]. In addition, where considerable heterogeneity was detected, supplementary sensitivity analyses were carried out by stratifying studies according to sample characteristics or treatment approaches to explore potential explanations for variability.

Risk of bias and certainty of evidence

The methodological quality of the included studies was assessed independently by two reviewers (R.L. and C.S.), with any disagreements resolved through consultation with the corresponding authors (Z.X. and P.Q.). Because most of the eligible studies were observational in nature, the risk of bias was evaluated using the Newcastle–Ottawa Scale (NOS), which considers aspects of participant selection, comparability of study groups, and outcome assessment [19]. For each study, scores were assigned across these domains, and an overall rating of methodological quality was determined. In addition, key features such as clarity of RTS definitions, completeness of follow-up, and transparency of reporting were examined to provide a more comprehensive appraisal of study validity.

The certainty of evidence for the two main outcomes (RTS rate and time to RTS) was assessed using the GRADE framework, which classifies evidence into four levels: high, moderate, low, and very low [20].

Statistical analysis

We performed a proportion-based meta-analysis to estimate pooled return-to-sport rates in elite athletes following ankle syndesmosis injuries. For each study, the RTS rate was calculated as the number of athletes who resumed sport participation divided by the total sample size. To stabilize variances, proportions were synthesized using logit transformation, and pooled estimates with 95% confidence intervals were generated. For studies that reported extreme event rates (0% or 100%), a continuity correction of 0.5 was applied to both the numerator and denominator to allow transformation [21].

Time to RTS was analyzed separately for studies reporting continuous data. When means and standard deviations (SDs) were provided, values were pooled using a random-effects model. If only median and range or interquartile range were available, these were converted into approximate means and SDs based on established statistical methods. In instances where measures of variability were absent, unweighted averages were calculated to provide a descriptive summary.

Between-study heterogeneity was quantified using the I^2 statistic and τ^2. When substantial heterogeneity was present ($I^2 > 50\%$), a random-effects model with restricted maximum likelihood estimation was employed [22]. Prespecified subgroup analyses were carried out to explore potential sources of heterogeneity, such as type of management (surgical vs conservative), fixation method, injury characteristics (isolated vs combined syndesmosis injury), and sport discipline. All statistical analyses were conducted using R (metafor and meta packages).

Results

Study selection and basic characteristics

A total of 1,138 records were retrieved from the database searches conducted up to September 16, 2025. Following the screening of titles and abstracts, 50 studies were considered potentially eligible. After full-text evaluation, 14 studies met the inclusion criteria and were incorporated into the meta-analysis. The final set of included studies comprised [7, 14, 15, 23–33]. The overall process of study identification, screening, eligibility assessment, and inclusion is detailed in the PRISMA flow diagram (Fig. 1).

Across the 14 included studies, a total of 901 elite athletes (mean age = 26.8 years) from six countries were examined. Return-to-sport outcomes following ankle syndesmosis injuries were assessed in all studies. For each investigation, the overall sample size together with the number of athletes who achieved RTS was reported. Treatment modalities such as suture-button fixation and nonoperative management were described in most studies, whereas several also provided information on the time to RTS following treatment. Detailed characteristics of the included studies are summarized in Table 1.

Risk of bias and publication bias

Overall, the risk of bias assessment using the Newcastle–Ottawa Scale (NOS) indicated that most included studies were of high to moderate quality (Fig. 2). The majority of studies scored well in the Selection and Outcome/Exposure domains, suggesting that study populations were representative and outcome assessment was adequately reported. Moderate quality ratings were primarily attributable to lower scores in the Comparability domain, reflecting limited adjustment for potential confounding factors. Only a few studies [14, 28] were rated as moderate quality due to concerns across multiple domains. Taken together, these findings suggest that while the overall methodological quality of the evidence base was acceptable, residual risks of bias remain—particularly regarding control of confounding—which should be considered when interpreting the pooled results.

Regarding publication bias, the funnel plot appeared asymmetric for RTS, and Egger's regression test confirmed evidence of small-study effects ($z = 3.79$, $p = 0.0002$), suggesting potential publication bias (Fig. 3). Visual inspection indicated that several small studies with high RTS estimates were clustered on the right side

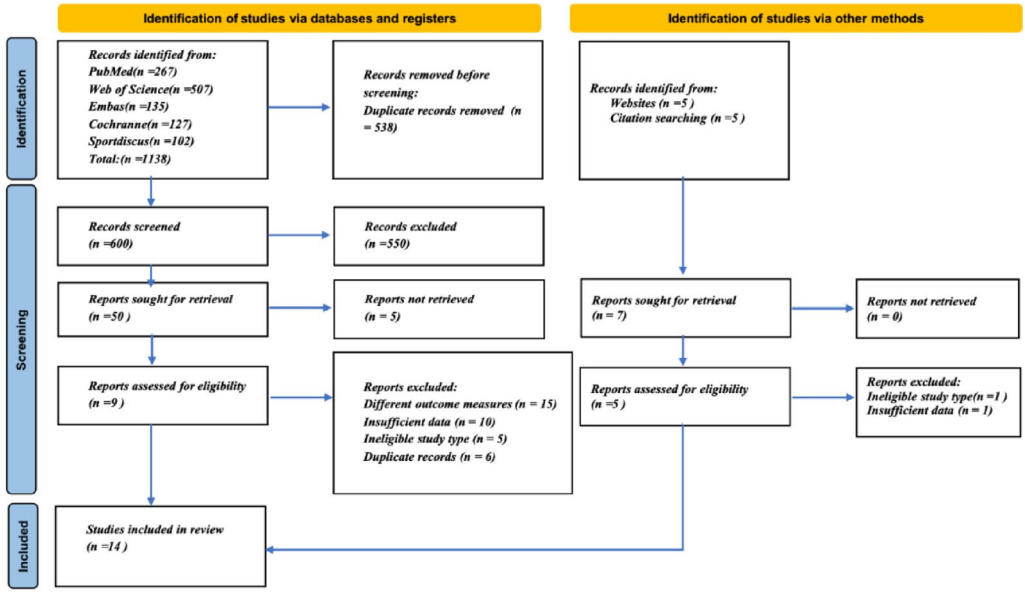


Fig. 1 PRISMA 2020 flow diagram showing the process of study identification, screening, eligibility assessment, and inclusion

Table 1 Characteristics of included literature

References	Country	N (total sample)	N (RTS)	Return to sports rate	Time to RTS (days)	Level of sport	RTS Definition	Injury grade	Treatment modality
D'Hooghe [15]	Qatar	110	110	100%	103.0	professional football players	Return to match play	West Point Grade IIB or III	Suture button fixation
Mollon [23]	Canada	19	19	100%	41.0	professional hockey players	Return to match play	MRI-confirmed anterior–inferior tibiofibular ligament disruption	Not reported
Latha [24]	UK	18	18	100%	64.0	professional rugby players	Return to match play	MRI-confirmed diastasis	Suture button fixation
Calde [7]	UK	28	28	100%	45.0	athletes	Return to match play	West Point Grade IIa	Nonoperative management
Sikka [25]	USA	36	36	100%	N/A	National Football League players	Return to match play	MRI-confirmed anterior talofibular ligament ligament injury	Nonoperative management
Howard [26]	USA	16	16	100%	30.0	National Football League players	Return to match play	Clinically stable syndesmosis sprain (consistent with West Point Grade I–IIa)	Nonoperative management
Wright [27]	USA	13	13	100%	38.0	professional hockey players	Return to match play	Clinically diagnosed stable syndesmosis sprain (no formal grade reported)	Nonoperative management
Miller [28]	USA	20	20	100%	15.5	collegiate football players	Return to match play	West Point Grade I	Nonoperative management
Taylor [29]	USA	6	6	100%	41.0	college athletes	Return to match play	West Point Grade III	Suture button fixation
Nussbaum [30]	USA	60	60	100%	13.4	college athletes	Return to match play	Clinically diagnosed stable syndesmosis sprain (no formal grade reported)	Nonoperative management
Kuhn [31]	USA	11	11	100%	30.0	National Football League players	Return to match play	Clinically diagnosed stable syndesmosis sprain (no formal grade reported)	Nonoperative management
DeFroda [14]	USA	533	478	89.7%	80.5	National Football League players	Return to match play	Clinically diagnosed stable syndesmosis sprain (no formal grade reported)	Not reported
Altmeppen [32]	Germany	20	20	100%	133	athletes	Return to match play	West Point Grade IIb or III	Suture button fixation
Wever [33]	South Africa	11	11	100%	120	professional rugby players	Return to match play	West Point Grade IIb or III	Suture button fixation

of the plot, whereas studies with lower RTS estimates were sparse on the left side, contributing to the asymmetry. This pattern implies that studies reporting lower return-to-sport rates may have been less likely to be published or included, potentially leading to an overestimation of the pooled RTS proportion.

Certainty of evidence and sensitivity analysis

According to the GRADE assessment, the certainty of evidence for the pooled RTS rate was judged to be low, reflecting the fact that all included studies were observational in design and that concerns about publication bias prevented any higher rating. The certainty of evidence for time to RTS was rated as very low, reflecting heterogeneous and incomplete reporting across studies and the lack of a pooled quantitative estimate.

Sensitivity analysis using a leave-one-out approach under a random-effects model demonstrated that the pooled RTS estimate was highly stable (0.95–0.97) when omitting each study in turn. Excluding a study [14] increased the pooled estimate to 0.98 (95% CI 0.95–0.99) and reduced heterogeneity to 0%, indicating that this study accounted for most of the between-study variability. Overall, these findings support the robustness of the primary meta-analytic result (pooled RTS = 0.96 [0.93–0.98]; $I^2 = 26.6\%$), as illustrated in Fig. 4.

Main effect results

Rate of return to sport

In total, fourteen studies comprising 901 athletes were included in the quantitative synthesis. When the data were pooled using a random-effects model, the overall return-to-sport rate following ankle syndesmosis injuries

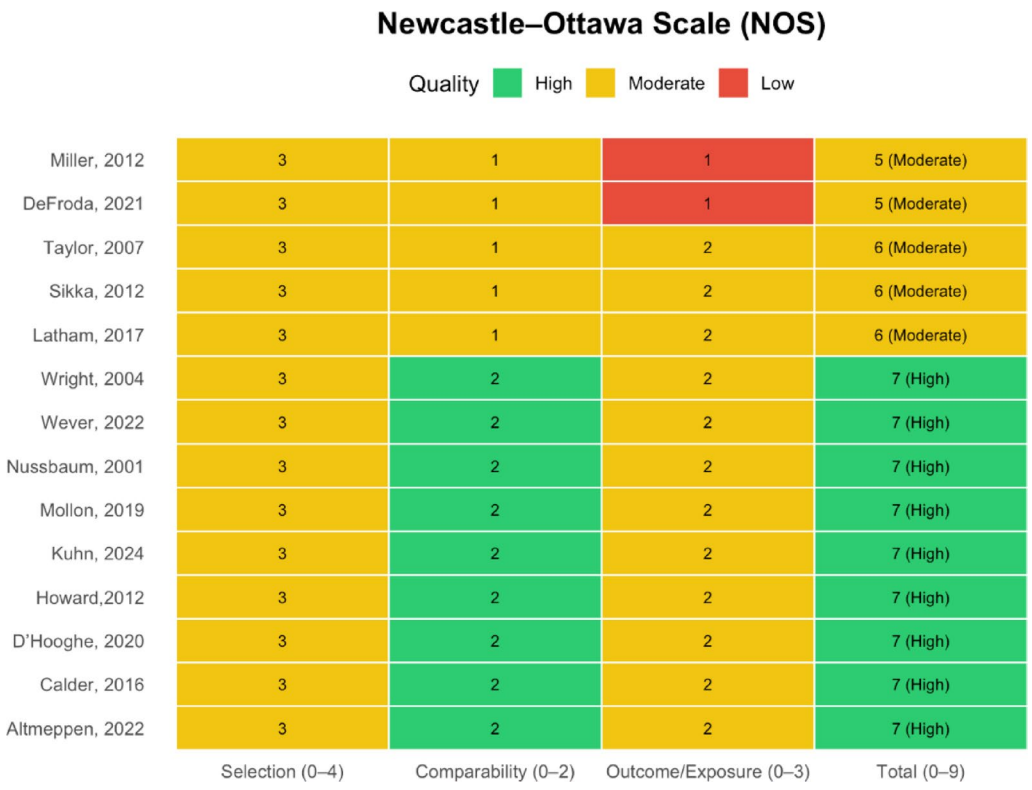


Fig. 2 Risk of bias assessment of the included studies using the Newcastle–Ottawa scale tool

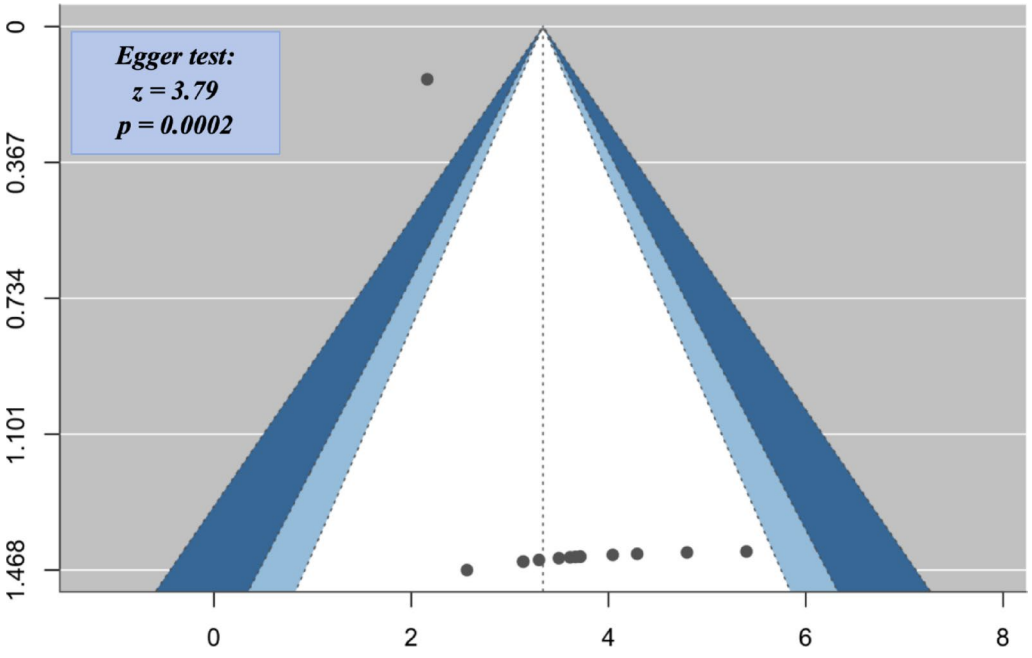


Fig. 3 Funnel plot for the rate of return to sport outcome showing evidence of publication bias

was calculated to be 96% (95% CI 93–98%), indicating that the vast majority of athletes were able to successfully resume sporting participation after such injuries (Fig. 5). Examination of between-study variability showed that heterogeneity was present at a low-to-moderate level ($I^2 = 26.6\%$, $\tau^2 = 0.3966$, $p = 0.169$). Although the magnitude of heterogeneity did not reach statistical significance, it nonetheless suggests that differences in study design, populations, or treatment modalities may have contributed to variation in the reported RTS rates. These

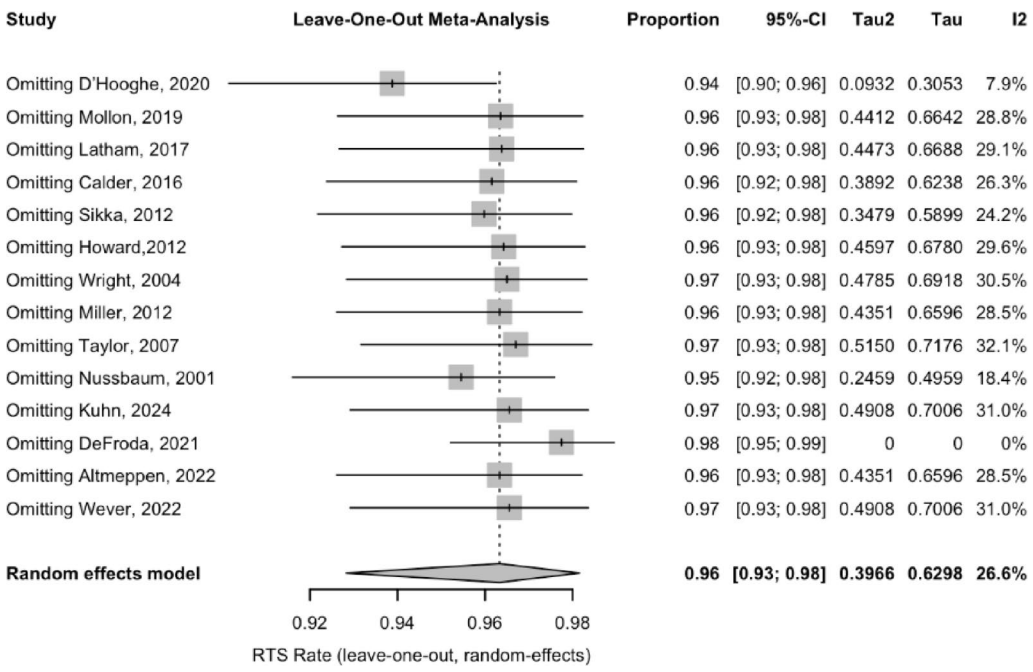


Fig. 4 Leave-one-out sensitivity analyses

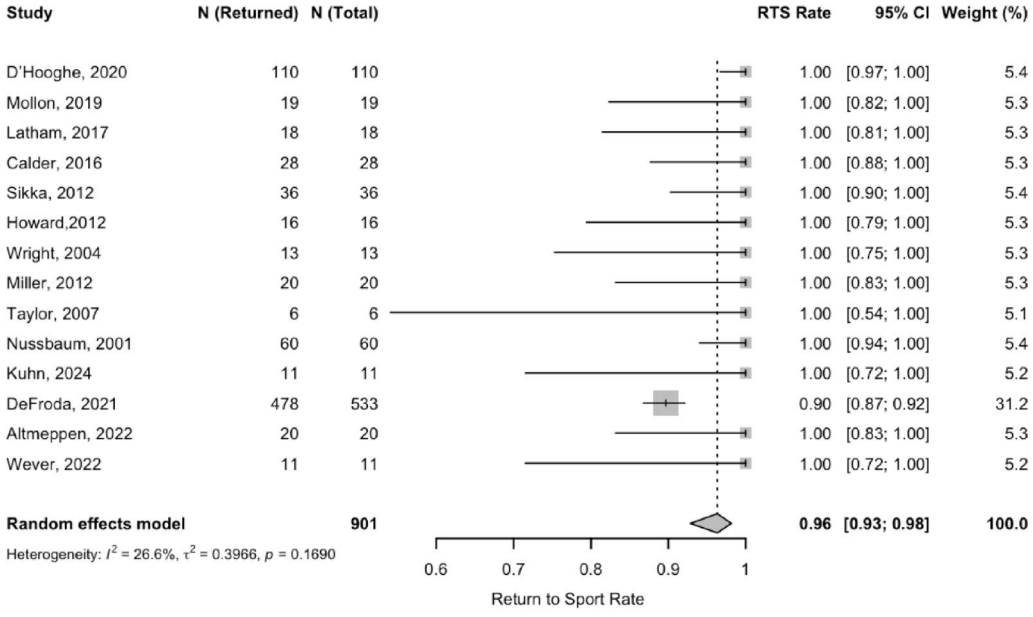


Fig. 5 Forest plot of the effect of the rate of return to sport (random-effects model)

findings highlight both the overall consistency of the evidence base and the need to consider potential contextual factors when interpreting the pooled effect.

To investigate possible contributors to heterogeneity, a subgroup analysis was carried out according to treatment modality (suture-button fixation versus nonoperative management). A pooled RTS rate of 98% (95% CI 92–99%; $I^2 = 0\%$) was observed among athletes treated with suture-button fixation, while a comparable estimate of 98% (95% CI 94–99%; $I^2 = 0\%$) was found for those managed nonoperatively. In contrast, studies in which treatment details were not explicitly reported yielded a lower pooled RTS rate of 90% (95% CI 83–95%). The test for subgroup differences was statistically significant ($\chi^2 = 8.59$, df = 2, $p = 0.0136$), indicating that variation in treatment modality and the adequacy of reporting were meaningful sources of between-study variability in RTS outcomes (Fig. 6).

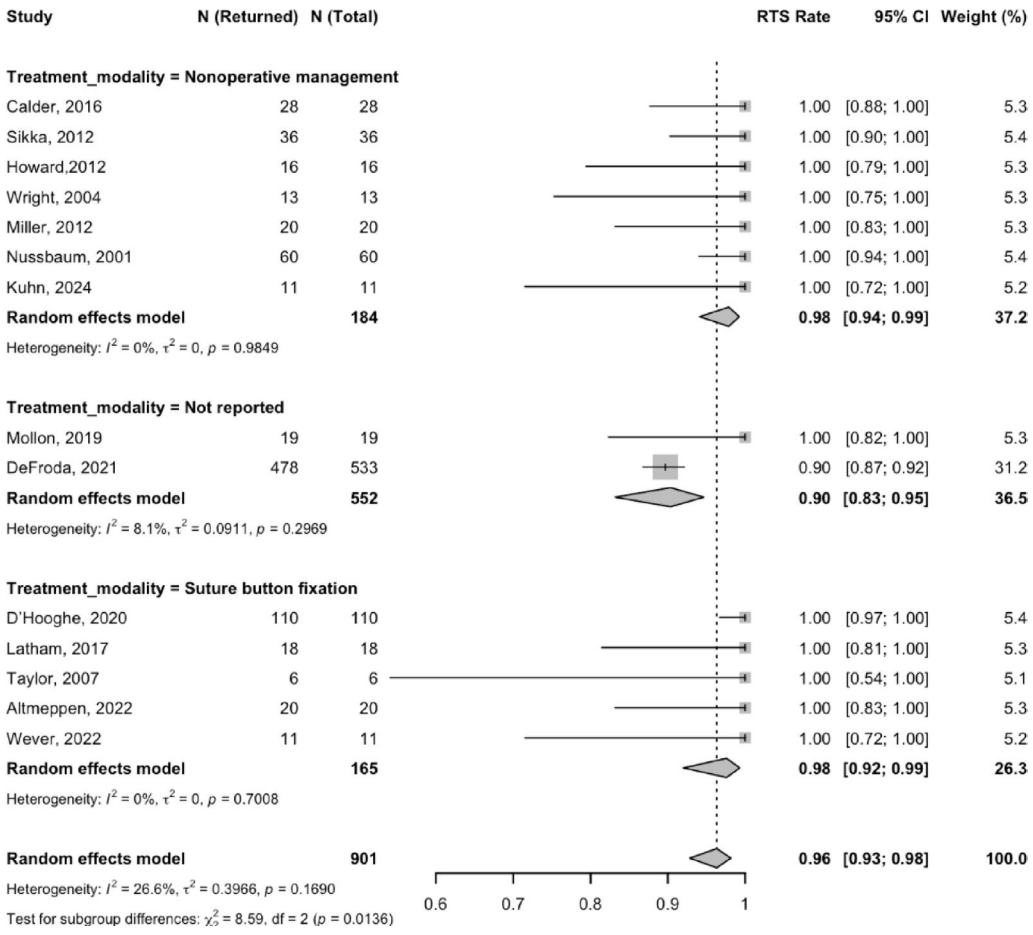


Fig. 6 Subgroup analysis of the rate of return to sport by treatment modality (suture-button fixation vs. nonoperative management)

Time to return to sport

Time to return to sport was reported in 13 of the included studies, although the format of reporting varied (mean values, medians, or single-point estimates), and most studies did not provide dispersion measures such as standard deviations, interquartile ranges, or ranges. Therefore, a formal meta-analysis of time outcomes could not be conducted. Instead, a simple descriptive approach was adopted, whereby the reported values were averaged across studies. The mean time to RTS across the available data was approximately 58 days (range: 13–133 days), indicating that elite athletes generally resumed sport within 2 months after injury. This estimate has limited inferential value and should be interpreted as a rough descriptive indication only, as it does not incorporate measures of variability, does not apply sample-size weighting, and does not account for differences in study design or treatment modality.

Discussion

This meta-analysis offers a comprehensive synthesis of current evidence on return-to-sport outcomes in elite athletes with ankle syndesmosis injuries. Fourteen studies involving 901 athletes were analyzed to estimate pooled RTS rates and average time to return, while also assessing heterogeneity and treatment-related effects. The overall RTS rate was high at 96%, with athletes resuming play after an average of 58 days. Outcomes were similarly favorable for those managed surgically with suture-button fixation and those treated conservatively. These findings provide updated insights into recovery trajectories following syndesmosis injuries in elite athletes and are further discussed below.

Rate of return to sport following syndesmotic injury

Our analysis demonstrated an overall return-to-sport rate of approximately 96% after ankle syndesmosis injuries, indicating that the vast majority of athletes are able to successfully resume sporting activity. This high success rate is consistent with prior literature. For example, a study [11] reported a 93.8% RTS to the pre-injury level of sport (and ~ 97.6% to any sport participation). Similarly, a recent meta-analysis focusing on elite athletes documented an RTS rate approaching 99% [13]. In a cohort of professional rugby players with Grade II syndesmosis sprains, all athletes returned to the same level of sport

following appropriate management [7]. Such consistency across studies underscores that, when treated properly, syndesmotic (high ankle) sprains generally have favorable outcomes in terms of eventual sport resumption.

In our subgroup analysis, we found no difference in RTS rates between treatment modalities: both those managed nonoperatively and those treated with suture-button fixation demonstrated rates of ~ 98%. This suggests that, when treatment selection is appropriate (operative for unstable injuries and conservative for stable injuries), athletes have an equally high likelihood of returning to sport. Taken together, these findings indicate that modern surgical stabilization techniques, particularly flexible suture-button devices, effectively restore ankle function, while less severe injuries treated nonoperatively also heal well, resulting in comparable outcomes in terms of athletes' ability to return to play.

Minor discrepancies among studies are likely attributable to important clinical factors such as injury grade, associated pathology, and type of sport. In our review, however, all studies defined RTS consistently as return to competitive match play, which helps to limit outcome-definition heterogeneity. Nonetheless, the overall consensus is that syndesmotic injuries, although potentially serious, rarely end athletic careers when managed appropriately.

Time to return to sport following syndesmotic injury

In our review, the mean time to RTS was approximately 58 days (about 8 weeks) following ankle syndesmosis injury. This finding is within the range reported in the literature, though individual studies vary depending on injury severity and the criteria used to define RTS. Our result is close to the ~ 52-day mean reported by Salameh [12], who similarly observed shorter RTS timelines after nonsurgical management and longer recovery after surgical fixation.

It is important to note that reported RTS timelines also vary according to the level of play and the definition of "return." Our ~ 8-week average likely represents return to full competition in a mixed population of athletes. More conservative protocols are often observed in elite sport. For instance, in a series of professional soccer players with surgically treated high-grade syndesmosis injuries, the first match appearance occurred at an average of 103 days (≈15 weeks) postoperatively, despite on-field rehabilitation beginning at around 5 weeks [15]. This extended time frame reflects the high physical demands of elite sport, where athletes are cleared for match play only when nearly fully recovered.

Overall, our finding of ~ 58 days to RTS is consistent with the general consensus that syndesmotic injuries typically sideline athletes for approximately 6–8 weeks. This is considerably longer than the downtime associated with common lateral ankle sprains (often 1–3 weeks) [34], underscoring the greater severity of high ankle sprains. Both our findings and the existing literature suggest that, with appropriate management, athletes can expect to return to sport within about two months, balancing sufficient healing time with the goal of achieving full functional recovery.

Implications for clinical practice and rehabilitation protocols

Although overall RTS rates were uniformly high, the choice between operative and nonoperative management should continue to be guided primarily by injury stability and severity. Stable, low-grade syndesmosis sprains (e.g., West Point Grade I–IIa) typically respond well to structured conservative rehabilitation, whereas unstable injuries (e.g., Grade IIb–III or MRI-confirmed diastasis) generally require surgical stabilization to restore mortise congruency and allow safe progression to sport-specific loading.

Rehabilitation timelines should likewise be individualized: while many athletes return within 6–8 weeks, those competing at the highest levels often follow a more conservative progression, incorporating objective criteria such as pain-free weightbearing, restoration of strength and proprioception, and satisfactory performance on functional and sport-specific tests before returning to full match play. These considerations underline the need for severity-based treatment pathways and criteria-led rehabilitation protocols to optimize recovery and minimize the risk of recurrent injury.

Comparison with previous systematic reviews

Compared with previous systematic reviews on RTS after ankle syndesmosis injuries [11–13], the present study offers several notable advances. First, it synthesizes the most recent evidence available up to 2025, incorporating studies published between 2022 and 2025 that were not captured in earlier reviews. Second, our review applied a more rigorous and standardized RTS definition by including only studies that reported return to competitive match play, thereby providing a more precise and performance-relevant outcome for elite athletes. Third, the methodological rigor of this review exceeds that of prior syntheses: we expanded the search across additional databases, implemented sensitivity analyses, and applied logit transformation to stabilize variance before generating pooled estimates with 95% confidence intervals. Collectively, these methodological and conceptual refinements position this study as an updated and more robust contribution to the current literature.

Limitations and future directions

This review has several limitations. First, although the pooled RTS rate was robust, reporting of time to RTS was inconsistent and often lacked measures of variability, which precluded a formal meta-analysis and limited us to a descriptive summary. Consequently, the estimated average time to RTS of 58 days has limited inferential value and should be interpreted only as a crude indication of the typical timeframe rather than a precise pooled estimate. Although more sophisticated approaches (e.g., converting medians to means or using sample-size–weighted averages) were considered, the lack of dispersion data and heterogeneous reporting of time to RTS made these methods infeasible. Second, most included studies were observational and retrospective in design, with moderate methodological quality and limited adjustment for confounding factors, which may affect the certainty of the findings. Third, evidence of publication bias suggests that the pooled RTS rate may be overestimated. Therefore, the pooled estimate should be interpreted with caution, and care should be taken when applying this value in clinical contexts. In addition, these methodological constraints, together with incomplete reporting of key clinical details, limit our ability to fully explore how clinical factors such as injury grade, associated pathology, and sport type influence RTS outcomes and contribute to between-study heterogeneity.

Future research should prioritize more rigorous prospective designs with standardized outcome reporting, including both RTS rates and measures of variability for return times. Further studies are also needed to clarify the influence of rehabilitation protocols and sport-specific demands on RTS and to provide higher-quality evidence to guide management in elite athletes.

Elite athletes sustaining ankle syndesmosis injuries demonstrate a high likelihood of returning to sport, with a pooled RTS rate of 96%. Comparable outcomes were observed following both suture-button fixation and non-operative management, with most athletes resuming play within 2 months. Most athletes can anticipate favorable outcomes regardless of treatment strategy, though management should still be individualized based on injury severity and sport-specific demands.

Conclusion

Elite athletes sustaining ankle syndesmosis injuries demonstrate a high likelihood of returning to sport, with a pooled RTS rate of 96%. Comparable outcomes were observed following both suture-button fixation and non-operative management, with most athletes resuming play within 2 months. Most athletes can anticipate favorable outcomes regardless of treatment strategy, though management should still be individualized based on injury severity and sport-specific demands.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13018-025-06566-6.

Supplementary Material 1

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Author contributions

Ran Li and Changlu Sun: contributed to the study design, performed the literature search, study selection, data extraction, and quality assessment, conducted the data analysis, created figures and tables, wrote and revised the initial drafts, and approved the final version of the article. Zhonghao Xu and Pengkun Qi: contributed to the study design, verified the extracted data and analyses, wrote and reviewed the initial drafts, and approved the final version of the article.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable. This study is a systematic review and meta-analysis of previously published research and does not involve human participants or animal subjects directly.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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